

DEVICE ANATOMY 2.0

Engineering Investigation of an Electronic Device at End-of-Life

Course	E-Waste & Environmental Management	Class	TE INFT
Duration	45-60 minutes	Marks	10

Group Details

Group No.:	03	Device Investigated:	Earphone
Date:	02/09/2026	Condition:	Non functional
Member 1:	Vaishnavi Bagade	Member 2:	Imran Shaikh
Member 3:	Amol Patil	Member 4:	Yash Jadhav
Member 5:	Malhaar Nikam		

Condition: ☐ Functional ☒ Non-functional ☐ Damaged ☐ Unknown

Activity Evidence

During the activity, photograph all group members actively participating in the activity. The photo should be taken from a distance where all group members are visible in the photo.



Photo caption: Our group investigating the anatomy and end-of-life characteristics of a earphone.

Group members visible: All group members visible
(Amol, Yash, Imran, Vaishnavi, Malhaar)

ACTIVITY CHALLENGE

You are an e-waste engineer. Your task is to investigate the material anatomy of an electronic product, identify recovery opportunities and environmental risks, and propose engineering changes that improve its end-of-life management.

Core question: Can you make this device more circular?

Part A — Device Profile

Device name	Earphone
Primary function	To listen audios, music etc.
Why does this device become e-waste?	☐ Obsolescence ☒ Damage ☐ Upgrade ☐ Repair not economical ☐ Failure ☐ Other: _____

Part B — Engineering Anatomy

Identify at least 6 major components. Do not merely name the component — explain the material choice.

#	Component	Function	Material(s)	Why is this material suitable?
1	Speaker/Driver	converts electrical signal to sound	Plastic/Polycarbonate	Lightweight, durable
2	Wire/cables	carry audio signal	copper	Good electrical conductivity.
3	Microphone	converts voice to electrical signal	Silicon + metal + plastic	ideal for usage
4	Control button	control play, pause	FR4 fiberglass + copper	strong, heat resistant & electrically insulating
5	Earphone mesh	protective mesh	stainless steel	strong, corrosion resistant & thin
6	Ear tips	protective part	silicon/rubber	soft flexible and comfortable

Engineering Challenge: Which component demonstrates the most interesting material selection? Why?

Speaker/driver because it combines a strong neodymium magnet, copper voice coil and lightweight diaphragm to convert electrical signal into sound in very small space.

Part C — Material Detective

Tick the major material categories present in your device:

☒ Ferrous metals ☐ Aluminium ☒ Copper ☒ Precious metals (silver)

☐ Engineering plastics ☐ Glass ☐ Ceramics ☐ Semiconductor materials

☒ Rubber/elastomers ☒ Other: magnet, carbon

	Material 1	Material 2
Name	copper	Neodymium Magnet
Where is it used?	Inside the wire	Inside the speaker
Why is it used?	Excellent electrical conductivity, i.e. efficiently carries audio signals	helps convert sound electrical signal to sound

	Material 1	Material 2
Can it be recovered/recycled? How?	yes can be separated and sent to specialized metal recycling	Yes can be separated and sent to magnet recycling.

Part D — Hidden Value: The Urban Mine

Identify the three most valuable/recoverable materials in your device.

Material	Component containing it	Potential value	Recovery difficulty
1. Copper	wires & voice coil	Low / Medium / High	Low / Medium / High
2. Neodymium	speaker / driver magnet	Low / Medium / High	Low / Medium / High
3. Gold silver / nickel plating	Audio jack / contacts	Low / Medium / High	Low / Medium / High

Critical Thinking: Is the most valuable material necessarily the easiest to recover? Explain.

No, A material can have high value but still be difficult to recover because it may be present in very small quantity or mixed with other material

Part E — Hazard Detective

Identify components that may require controlled handling.

Component	Possible hazard	Risk if improperly disposed	Handling requirement
Speaker	magnet / metal parts	Environmental contamination	sp separate & send for ewaste recycling
PCB	Electronic components	Release harmful substance	Handle as ewaste
Wires	pvc/TPE insulation	Toxic fumes	Do not burn. send for cable recycling
3.5 mm Jack	metal / plating	metal waste	separate & recycle

Most environmentally concerning component: PCB

Why? contains electronic components, copper & solder that should not be disposed of with normal household waste. improper processing can release harmful substance and also waste recoverable materials

Part F — End-of-Life Decision

Recommend the most appropriate pathway for each major component.

Component	Reuse	Repair	Refurbish	Recycle	Controlled Disposal
	☐	☒	☐	☐	☐
	☐	☒	☐	☐	☐
	☐	☐	☒	☐	☐
	☒	☐	☐	☐	☐

Justify ONE of your decisions:

wire/cable should be repaired or reused whenever possible because cable damage is common and repairing it can extend the earphone's life while reducing e-waste

Part G — Design for Disassembly

Imagine that you are the engineer who designed this product. Identify three design features that make recycling, repair or disassembly difficult.

#	Existing design problem
Problem 1	Earphone housing is glued or tightly sealed making it difficult to open without damaging
Problem 2	Very thin wires are soldered directly to speaker making replacement and repair difficult
Problem 3	Different materials such as plastic, metal and electronics are combined in a small assembly, making separation difficult

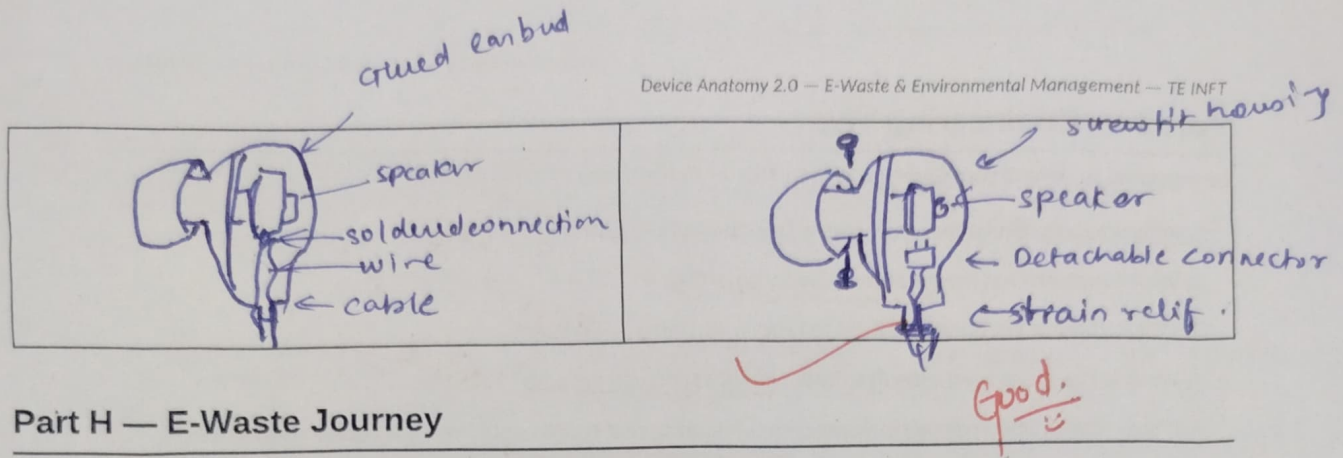
Now redesign it!

Propose three engineering modifications that would make the product more circular.

Existing Design Problem	Proposed Design Change	Sustainability Benefit
Existing housing is difficult to open	use small screw	Easier repair
wires are directly soldered	use small detachable connector	Easy replacement of parts
Different materials are difficult to separate	use clearly separated material module	easier recycling

Sketch your redesigned device/component — show at least one change that improves repair, disassembly or material recovery.

ORIGINAL	REDESIGNED
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Part H — E-Waste Journey

Trace the end-of-life journey of ONE component.

Component selected: copper wire / cable

DEVICE	CONSUMER DISCARD	COLLECTION	DISMANTLING	MATERIAL RECOVERY	NEW PRODUCT / RESOURCE
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What could go wrong at any stage?

It earphone is thrown into general waste, valuable copper may be lost. Improper dismantling or burning of insulated wires can also cause environmental pollution and release harmful fumes.

Part I — Circular Economy Decision

Our device should NOT simply be treated as waste because...

It contains reusable components and valuable materials such as copper, magnet, plastic, and electronics part that can be repaired, reused or recycle.

Which circular strategy would have the greatest impact?

☒ Repair ☐ Reuse ☐ Refurbishment ☐ Remanufacturing

☐ Recycling ☐ Material recovery

Why?

Repairing the earphone extends its useful life and prevents the entire device from becoming e-waste. It also requires fewer new raw materials.

Part J — Engineering Viva

Be prepared to defend your answers. Your instructor may ask any of the following:

1. Why was a particular material selected for one component? *Cu*
2. Which material has the greatest recovery potential?
3. Which component is most difficult to recycle and why? *PCB*
4. What is the single most effective design modification you propose? *Nice*
5. If your sustainable redesign increases manufacturing cost by 8%, would you still recommend it? Why?

Final Group Verdict

The most valuable component is:	<i>copper containing wires</i>
The component with the greatest environmental concern is:	<i>control PCB</i>
The biggest design-for-recycling problem is:	<i>Glued / sealed housing & difficult component separation</i>
Our most important redesign recommendation is:	<i>Use screw / snap fit housing & detachable connections</i>
ONE-SENTENCE ENGINEERING VERDICT We recommend <i>designing earphone with easily removable & recyclable components</i> because <i>this will extend its life, simplify repair</i>	

Assessment Rubric — 10 Marks

Criterion	Excellent (2)	Good (1.5)	Developing (1)	Marks
1. Engineering Anatomy	Accurate component + function + material analysis	Mostly accurate	Basic identification	/2
2. Material & Recovery Analysis	Strong material/recovery reasoning	Adequate reasoning	Limited reasoning	/2
3. Hazard & E-waste Assessment	Identifies risks with clear justification	Identifies major risks	Basic identification	/2
4. Design for Disassembly	Innovative, feasible engineering solutions	Relevant solutions	Basic suggestions	/2
5. Engineering Defence/Viva	Clear, evidence-based justification	Reasonable explanation	Limited justification	/2
TOTAL				10

Reflection

What surprised you most about the materials hidden inside your device?

Simple earbuds contains many different materials and work together to make earphone small, lightweight & functional

After this activity, would you still call this device "waste"? Why?

No, I would not simply call it waste because it contains valuable and reusable material such as copper & magnet. It can be repaired, reused, recycled.

KEY TAKEAWAY

E-waste is not merely waste — it is a complex mixture of materials, components, resources and risks that requires engineering solutions for responsible recovery.